Responsible Data Analytics Report
Urban Sustainability & Renewable Energy Potential in the Baltic States

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***Abstract*— We present a four-phase analytics framework to understand and harness renewable energy potential in the Baltic capitals—Vilnius, Tallinn, and Riga—by linking local weather variability to urban sustainability. In the descriptive phase, we map multi-year trends and seasonality in solar radiation, temperature, wind, and humidity, revealing characteristic summer peaks and winter lulls. The diagnostic analysis quantifies interdependencies—e.g. how near-surface humidity and surface temperature jointly modulate clear-sky radiation—which informs when and where renewables perform best. Moving to predictive analytics, we build a time-aware stacking ensemble (random forest, gradient boosting, SVR, MLP with a RidgeCV meta-learner) that, after enforcing a full-year rolling training window, forecasts daily solar irradiance with test RMSEs of 0.58–0.66 kW·h/m²/day ($R^2 \approx 0.93$). Finally, our prescriptive scenarios translate these forecasts into actionable strategies—such as adjusting buffer storage or dynamically curtailing other generators during expected low-insolation spells—to optimize grid integration. Together, these insights equip city planners and utilities with a data-driven playbook for climate-resilient, renewable-rich urban energy systems in northern Europe.**

Keywords*— *Solar Radiation, Data Analytics, Sustainability, Climate Change, Machine Learning, Meteorological Data

INTRODUCTION

As the climate crisis intensifies, renewable energy has become vital to sustainable development. The European Union (EU) aims to reduce CO_2 emissions, boost energy efficiency, and achieve at least 42% renewable energy in its overall mix (Renewable Energy Targets, n.d.)[1]. Meeting these goals requires not only innovation but also strategic urban planning, especially in cities, major centers of energy use and emissions.

Urban integration of renewable energy depends heavily on local meteorological conditions, which directly impact the performance of solar, wind, and other weather-dependent systems. This is especially relevant in the Baltic States; Lithuania, Latvia, and Estonia where Vilnius, Riga, and Tallinn are navigating the transition to low-carbon energy while adapting to growing energy demands and climate risks.

Despite access to rich meteorological data, there's limited applied research connecting weather variability to renewable energy output and urban sustainability in this region. This project addresses that gap by analyzing how environmental factors like solar radiation, temperature, wind speed, and humidity affect renewable energy generation and urban resilience.

Leveraging five years of meteorological data from NASA's POWER platform, the study applies responsible data analytics techniques to uncover both historical trends and predictive insights. The primary objectives are to:

1. Analyze how key weather parameters affect renewable energy potential and urban sustainability in the three capitals.

2. Provide actionable, evidence-based recommendations for energy strategies and urban planning that are tailored to regional climatic conditions and aligned with EU climate policy.

Using a combination of descriptive, diagnostic, and predictive analytics, this project ultimately seeks to translate data-driven insights into prescriptive guidance for policymakers and city planners. The overarching goal is to support sustainable urban transitions and contribute to the broader European effort toward a climate-resilient, low-carbon future.

PROBLEM STATEMENT

To contribute to the EU's renewable energy transition, this project investigates how local meteorological conditions influence renewable energy potential in the Baltic capitals; Vilnius, Tallinn, and Riga. Despite the availability of relevant climate data, there is limited applied research linking weather variability to renewable energy performance and urban sustainability in Vilnius, Tallinn, and Riga. This project addresses that gap by investigating how meteorological parameters affect renewable energy potential in these cities. With our main research question being:

How do variations in meteorological parameters influence renewable energy potential and urban sustainability in Vilnius, Tallinn, and Riga?

To address this question comprehensively, we break it down into four sub-questions aligned with the key phases of data analytics:

1. Descriptive Analytics: What are the historical trends and seasonal patterns of key meteorological variables such as solar radiation, temperature, wind, and humidity?

2. Diagnostic Analytics: How do solar, temperature, moisture, and wind variables correlate and interact, and what implications do these relationships have for renewable energy generation?

3. Predictive Analytics: Can we develop models to predict solar radiation based on other meteorological data that impacts renewable energy generation and potential?

4. Prescriptive Analytics: What general strategies can be used to optimize renewable energy deployment, and what specific actions can be taken based on predicted solar radiation?

By systematically addressing each of these sub-questions, we aim to generate actionable insights that support responsible, climate-conscious energy planning in the Baltic capitals.

Data Description

For this project we utilize five years of meteorological data (2018–2022) from NASA's POWER platform, focusing on the Baltic capitals: Vilnius, Riga, and Tallinn. These cities were selected due to their increasing importance in the European Union's green energy transition and their exposure to regional climate variability. The dataset includes key weather parameters related to solar radiation, temperature, humidity, wind, and precipitation, forming the basis for the descriptive, diagnostic, predictive, and prescriptive phases of our analysis.

1.Data preprocessing:

To ensure a clean and consistent foundation for analysis, we began by loading the CSV files. Each dataset was examined to verify its structure and format, ensuring that the same variables were present and consistently labeled across all cities. Once validated, the datasets were merged into a single, unified dataframe to facilitate comparative analysis across locations. We also unified and converted the "YEAR", "Mo", and "DY" columns to datetime and Set 'DATE' as the index to have a time series data format to leverage this characteristics of our data. We performed checks for missing values, duplicates, and inconsistencies, confirming that the data was complete and free from redundancies. Data types: All columns were of type float64 (numerical), with the exception of the "City" column, which remains as an object type for categorical identification. This variable will be encoded (e.g., via one-hot encoding) at a later stage during the predictive modeling process. Key preprocessing steps included:

- Standardizing column names (e.g., ensuring uniform capitalization in date fields like "year")
- Checking for missing values: No missing values were found across any rows or columns.
- Duplicate detection: No duplicate entries were present, indicating strong data integrity.

We also performed outlier analysis, see Fig. 1 (The variable WD10M (wind direction at 10 meters) exhibited a broad range of values that visually overwhelmed other variables and since it did not contain any outliers, it is excluded from the plot).

In order to better understand the features that have outliers and to confirm outliers numerically, we double checked them and after assessing they were determined to be genuine observations rather than anomalies caused by data entry errors or noise. As such, these values will be retained during the initial stages of analysis to preserve the integrity and representativeness of the data. If necessary, transformations will be applied in later stages to manage their influence. In particular, when developing machine learning models, techniques such as StandardScaler or RobustScaler from the sklearn.preprocessing module will be considered to minimize the impact of outliers while maintaining the overall structure and variability of the dataset.

After this process, we also identified potential sources of bias and risk, such as the use of satellite-derived data that may not capture urban microclimate variations accurately. These considerations were noted to ensure a responsible and transparent data analytics process moving forward.

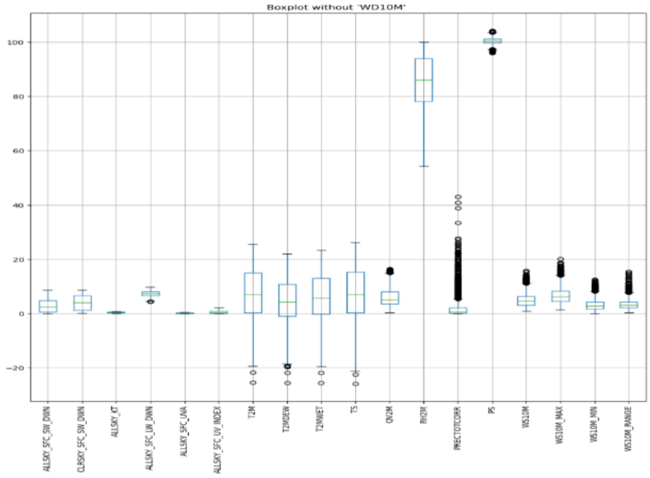

Fig. 1. Box plot of selected meteorological variables (excluding WD10M) for outlier analysis.

Case Sensitivity and Considerations

The focus on Vilnius, Tallinn, and Riga is both timely and socially significant. These cities are under pressure to align with the EU's 42.5% renewable energy target by 2030, while also navigating challenges related to urban growth, infrastructure modernization, and climate resilience. However, several important limitations must be acknowledged:

- The dataset includes only meteorological variables, meaning it does not reflect socioeconomic, infrastructural, or policy-related factors that are crucial for defining true urban sustainability.
- Insights drawn from just three cities may not generalize across regions with different climates or urban forms, posing a geographic bias.
- The dataset spans only five years, creating a short-term analysis bias and missing longer-term climate trends.
- Urban microclimates (e.g., heat islands) may not be accurately captured by satellite-derived data, raising positionality concerns about treating this data as purely objective.

These limitations highlight the importance of transparency in framing, ensuring that the insights generated are understood within the boundaries of what the dataset can and cannot represent.

Features(columns)	Category	Non-Technical description	Technical description
ALLSKY_SFC_SW_DWN	Solar Radiation & UV (CERES SYN1deg)	Total shortwave (SW) radiation received at the surface under all sky conditions (incl. clouds).	All Sky Surface Shortwave Downward Irradiance (kW-hr/m^2/day)
CLRSKY_SFC_SW_DWN	Solar Radiation & UV (CERES SYN1deg)	Shortwave radiation received at the surface assuming clear skies (no clouds).	Clear Sky Surface Shortwave Downward Irradiance (kW-hr/m^2/day)
ALLSKY_KT	Solar Radiation & UV (CERES SYN1deg)	Clearness index; fraction of solar radiation transmitted through the atmosphere.	All Sky Insolation Clearness Index (dimensionless)
ALLSKY_SFC_LW_DWN	Solar Radiation & UV (CERES SYN1deg)	Longwave (LW) radiation received at the surface under all sky conditions.	All Sky Surface Longwave Downward Irradiance (kW-hr/m^2/day)
ALLSKY_SFC_UVA	Solar Radiation & UV (CERES SYN1deg)	Ultraviolet-A (UVA) radiation received at the surface.	All Sky Surface UVA Irradiance (kW-hr/m^2/day)
ALLSKY_SFC_UV_INDEX	Solar Radiation & UV (CERES SYN1deg)	A measure of UV intensity, relevant for skin exposure risks.	All Sky Surface UV Index (W m-2 x 40)
T2M	Temperature & Humidity (MERRA-2)	Air temperature at 2 meters above the ground.	Temperature at 2 Meters (C)
T2MDEW	Temperature & Humidity (MERRA-2)	Dew point temperature, indicating saturation point of air moisture.	Dew/Frost Point at 2 Meters (C)
T2MWET	Temperature & Humidity (MERRA-2)	Wet bulb temperature, relevant for heat stress analysis.	Wet Bulb Temperature at 2 Meters (C)
TS	Temperature & Humidity (MERRA-2)	Skin temperature of the Earth’s surface (land/sea surface).	Earth Skin Temperature (C)
QV2M	Temperature & Humidity (MERRA-2)	Specific humidity; amount of water vapor in the air.	Specific Humidity at 2 Meters (g/kg)
RH2M	Temperature & Humidity (MERRA-2)	Relative humidity; moisture content relative to the air's capacity.	Relative Humidity at 2 Meters (%)
PRECTOTCORR	Precipitation & Pressure (MERRA-2)	Daily corrected total precipitation.	Precipitation Corrected (mm/day)
PS	Precipitation & Pressure (MERRA-2)	Surface pressure, useful in weather pattern analysis.	Surface Pressure (kPa)
WS10M	Wind (MERRA-2)	Wind speed at 10 meters above the ground.	Wind Speed at 10 Meters (m/s)
WS10M_MAX	Wind (MERRA-2)	Maximum wind speed recorded at 10 meters.	Wind Speed at 10 Meters Maximum (m/s)
WS10M_MIN	Wind (MERRA-2)	Minimum wind speed recorded at 10 meters.	Wind Speed at 10 Meters Minimum (m/s)
WS10M_RANGE	Wind (MERRA-2)	Difference between maximum and minimum wind speed.	Wind Speed at 10 Meters Range (m/s)
WD10M	Wind (MERRA-2)	Wind direction at 10 meters (in degrees).	Wind Direction at 10 Meters (Degrees)

Table. 1. NASA POWER features along their technical and non-technical definitions

Exploration of the Dataset and Relevant Variables

Our data set, the NASA POWER dataset, has a variety of meteorological variables and solar radiation measurements that are relevant to urban sustainability and renewable energy. It is comprised of 19 different features and (see Table. 1)you can see all the variables along their technical and non-technical definitions and their scales.

These variables collectively enable an in-depth exploration across descriptive, diagnostic, predictive, and prescriptive analyses related to urban energy and climate challenges.

Contextual Relevance of key parameters selection: Renewable Energy Systems and Urban Sustainability

Several studies show that only a few key meteorological variables consistently dominate both renewable energy potential estimation and urban sustainability analyses. For example, in the solar energy domain, the surface shortwave downward irradiance variables—both all-sky (ALLSKY_SFC_SW_DWN) and clear-sky (CLRSKY_SFC_SW_DWN)—along with the insolation clearness index (ALLSKY_KT), are fundamental because they directly determine the available solar energy for photovoltaic (PV) conversion.

Similarly, wind energy assessments rely heavily on wind speed metrics. Variables such as WS10M (including its maximum, minimum, and range) and wind direction (WD10M) critically influence turbine performance by determining the energy capture potential and the dynamics of the wake effects.

In urban sustainability studies, variables governing the urban microclimate and building energy demands—such as the 2-meter temperature (T2M), relative humidity (RH2M), and precipitation (PRECTOTCORR)—play a central role. These parameters not only affect the efficiency and output of renewable energy systems (for example, through their influence on PV and wind performance) but also drive urban energy consumption patterns by influencing heating, cooling, and ventilation needs, as well as indoor and outdoor air quality.

For example, Qu et al. (2025)[2] demonstrate the dominant influence of solar irradiance (as captured by variables like ALLSKY_SFC_SW_DWN and ALLSKY_KT) on energy consumption variability in renewable-dominated systems. Likewise, Alskaif et al. (2020)[3] provide a systematic analysis on how various meteorological variables affect PV output power estimation, reinforcing the importance of solar irradiance measurements alongside temperature and wind parameters. In addition,literature reviews by Rocha et al. (2019)[4] and Maduabuchi et al. (2023)[5] highlight that, for renewable energy potential forecasting, key weather parameters—particularly temperature, humidity, wind speed, and precipitation—are often sufficient to capture most of the variability needed for robust estimates.

Thus, while all 19 variables can contribute information, the most critical for renewable energy and urban sustainability are:

- Solar-related variables: ALLSKY_SFC_SW_DWN, CLRSKY_SFC_SW_DWN, and ALLSKY_KT
- Wind-related variables: WS10M (and its max/min/range) and WD10M
- Urban climate and energy demand variables: T2M, RH2M, and PRECTOTCORR

This focused subset not only drives the performance of renewable energy systems but also underpins urban sustainability by shaping local microclimates and energy consumption patterns.

To conclude the dataset sourced from NASA's POWER platform provides a comprehensive range of meteorological variables such as solar radiation, wind speed, temperature, and humidity which are essential for evaluating renewable energy potential in Vilnius, Riga, and Tallinn. The data is sufficiently robust to address the primary research question, and we intend to utilize the full dataset, covering five years of daily data for all three cities. However, to enhance the prescriptive analytics phase and link environmental patterns to urban sustainability, additional variables such as urban infrastructure data, energy consumption statistics, and policy-related information could be beneficial. Insights from the diagnostic phase have highlighted strong seasonal patterns and variable correlations that may impact model performance, particularly in forecasting and planning tasks. Furthermore, scientific literature underscores the importance of incorporating regional climate trends and socioeconomic context when evaluating renewable energy feasibility, suggesting that supplementing our current dataset with policy and urban planning data would strengthen the relevance and applicability of our findings.

What are the historical trends and seasonal patterns of key weather variables?

After preprocessing, ensuring the quality, understanding the features of our data, We can start answering our sub questions. we will be using a descriptive analysis to find out the question 'what happened'.

2. Descriptive Analytics

Once the problem had been clearly defined, we proceeded with a descriptive analysis of the available dataset to uncover patterns and extract meaningful insights. Basically, we want to answer the following question:

-What are the historical trends and seasonal patterns of key weather variables?

The main objective of this phase was to explore and summarize the historical meteorological data collected over five years for Vilnius, Riga, and Tallinn, using a responsible data analytics approach.

Exploratory Data Analysis (EDA)

To begin our analysis, we conducted an exploratory data analysis (EDA) to gain a comprehensive understanding of the dataset's structure and key features. We examined trends, seasonality, and distribution patterns across the key meteorological variables that we selected in the previous stage, including solar radiation, air temperature, wind speed, humidity, and precipitation all of which directly influence the feasibility and performance of renewable energy systems in urban environments.

T2M: Fig.2 visualizes the historical air temperature at 2 meters (T2M) for Vilnius, Tallinn, and Riga from 2018 to 2023 using a combination of line plots, bubble plots, and box plots. The top-left line chart reveals clear seasonal patterns, with temperatures rising during summer and dropping sharply in winter across all three cities, reflecting their similar continental climates. The bubble plot below it highlights the intensity of temperature variations, with larger bubbles marking more extreme values. On the right, the box plots summarize the overall temperature distributions for each city, showing that all three experienced similar ranges and medians, with Riga having a slightly warmer distribution on average. This comprehensive visualization effectively captures both the temporal and statistical aspects of temperature trends in the Baltic capitals.

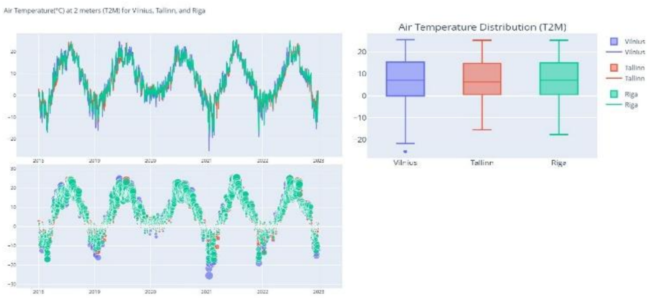

Fig. 2. Historical air temperature at 2 meters (T2M)

Relative Humidity at 2 meters (RH2M): Fig. 3 displays the relative humidity (RH2M) trends and distributions for Vilnius, Tallinn, and Riga from 2018 to 2023. The scatter plot on the left shows seasonal fluctuations, with humidity generally higher in colder months and lower in warmer ones. All three cities exhibit similar humidity patterns over time. The violin plots on the right summarize the distribution, showing that most humidity values fall between 70% and 100%, with Tallinn and Riga having slightly higher medians and more concentrated distributions than Vilnius. Overall, the visualization highlights consistent and humid atmospheric conditions across the three Baltic capitals.

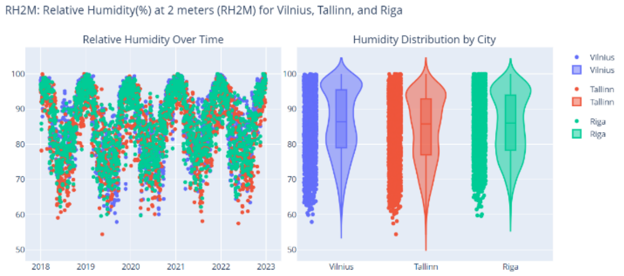

Fig. 3. The relative humidity (RH2M) trends and distributions

Daily total precipitation (PRECTOTCORR): Fig. 4 presents daily total precipitation (PRECTOTCORR) for Vilnius, Tallinn, and Riga from 2018 to 2023 using three views. The top plot shows non-zero precipitation values on a log scale, highlighting both frequent light rain and occasional heavy rainfall events. The bottom-left box plot compares precipitation distributions, showing that Riga tends to experience slightly higher extremes. The bottom-right linear scatter plot includes all values, emphasizing that most days in all three cities have low or no rainfall, with few intense outliers. Together, the graphs reveal that precipitation is infrequent, skewed, and generally similar across the three Baltic capitals.

Fig. 4. Daily total precipitation (PRECTOTCORR)

Wind-related variables (WS10M (and its max/min/range) and WD10M): Fig.5 presents a polar bar chart visualization of wind direction, frequency, and average speed intensity at 10 meters above ground level for Vilnius, Riga, and Tallinn. Each subplot represents one city, with bars indicating how frequently wind comes from different directions, while color intensity reflects the average wind speed based on categorized speed bins. The direction is displayed in degrees, rotating clockwise from 0° (north). Across all three cities, the dominant wind directions appear between 180° and 270° (southern and western winds), with Riga showing particularly frequent and stronger winds from the southwest. The colorbar on the right links color shades to wind speed categories, ranging from calm (1–2 m/s) to extreme (>15 m/s). This visualization effectively summarizes the prevailing wind patterns and their intensities in the three Baltic capitals.

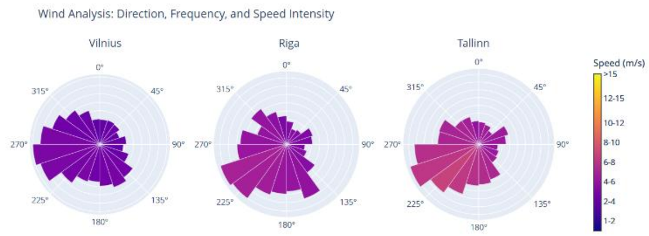

Fig. 5. Wind direction, frequency, and average speed intensity at 10 meters above ground level

Solar radiation(shortwave downward irradiance (ALLSKY_SFC_SW_DWN)): Fig. 6 illustrates the daily surface shortwave downward irradiance (ALLSKY_SFC_SW_DWN) in kilowatt-hours per square meter per day (kW·hr/m²/day) from 2018 to 2022 for Vilnius, Tallinn, and Riga. The data shows a clear seasonal pattern, with radiation levels peaking around June and July each year—indicating the highest solar energy potential during summer—and dropping to near zero during the winter months, reflecting the low sun angle and shorter daylight hours in these northern cities. All three cities follow a remarkably similar trend, highlighting the region's homogeneous solar climate, making this data valuable for understanding and optimizing solar energy utilization in the Baltic states.

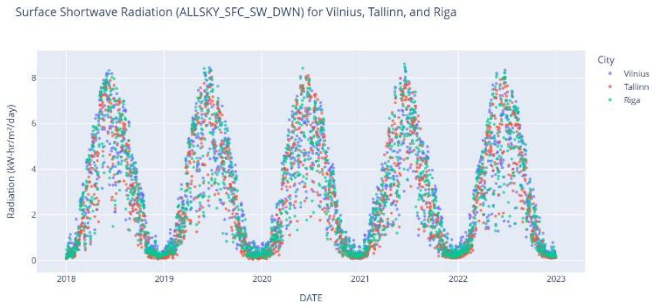

Fig. 6. Daily surface shortwave downward irradiance (ALLSKY_SFC_SW_DWN)

Fig. 7 illustrates the monthly distribution of solar radiation for Vilnius, Riga, and Tallinn, using a hierarchical view from city to month. The size of each outer segment represents the total solar radiation received in that month, while the color intensity (with red indicating higher daily averages) shows the average daily solar radiation (kW-hr/m²). Across all three cities, June consistently exhibits the highest solar intensity, followed by July and May, as shown by the deep red segments. In contrast, the winter months are represented in shades of blue, indicating minimal solar energy. This visualization highlights strong seasonal variation in solar energy availability across the Baltic capitals.

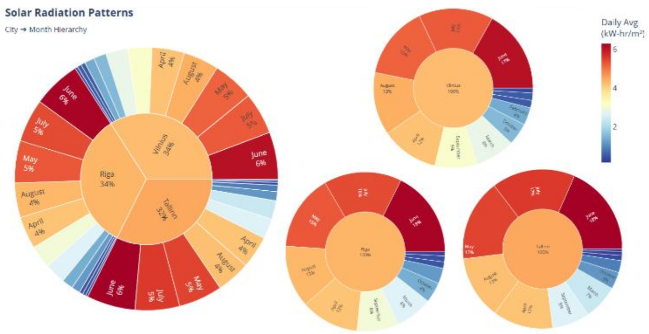

Fig. 7. Monthly distribution of solar radiation

Surface pressure (PS): Fig. 8 illustrates the surface pressure (PS) data for Vilnius, Tallinn, and Riga using two synchronized subplots. The left subplot is a time series scatter plot showing daily surface pressure (in kilopascals) from 2018 to 2022, where each dot represents a city's daily value. It reveals natural fluctuations in pressure, typically ranging between 98 and 102 kPa, with occasional outliers. The right subplot is a box plot summarizing the pressure distribution by city, highlighting that Tallinn and Riga generally experience slightly higher median pressures compared to Vilnius, and that Tallinn also shows the widest variation. Overall, this visualization provides insight into both the temporal trends and statistical spread of atmospheric pressure across the three Baltic capitals.

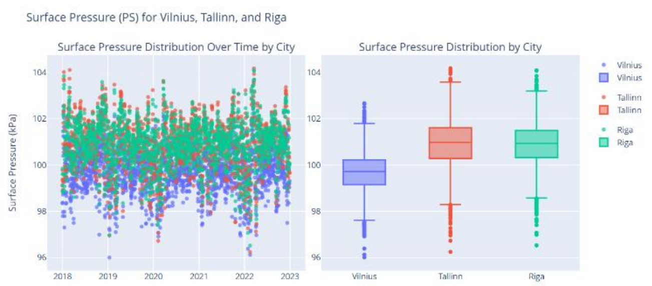

Fig. 8. The surface pressure (PS)

Statistical Distributions

We analyzed the statistical distributions of the most relevant variables to assess skewness, outliers, and variability. This helped us identify which features were normally distributed and which displayed significant deviations or extreme values.

Variable Properties

The dataset consists primarily of numerical variables, with the exception of a categorical "City" variable, which was used to differentiate between the three locations. Each numerical feature was checked for consistency in units and scale. The absence of missing values or duplicates further supported the reliability of the data.

Data Limitations and Challenges

While the dataset is comprehensive from a meteorological perspective, it lacks urban, socioeconomic, and infrastructure-related features that are critical for a holistic understanding of urban sustainability. This limits the scope of prescriptive insights unless supplemented with external data sources. Additionally, the use of satellite-derived data introduces a layer of generalization that may not fully capture microclimates or city-specific urban effects.

Responsible Framing of Data

We recognize that relying solely on meteorological data to assess "urban sustainability" introduces a narrow framing of the problem. Sustainability also encompasses social equity, economic accessibility, and governance all of which are absent from the dataset. By acknowledging this, we aim to be transparent about the scope and limitations of our descriptive analysis and to frame our insights as part of a larger, multi-dimensional context.

To conclude the descriptive analysis of meteorological data from 2018 to 2023 for Vilnius, Riga, and Tallinn reveals consistent and regionally coherent weather patterns across the Baltic capitals, with distinct seasonal trends that hold critical implications for renewable energy planning.

Air temperature (T2M) exhibits a clear seasonal cycle, peaking in July and reaching its lowest in January. All three cities demonstrate similar temperature behaviors, indicative of their shared continental climate, with Riga slightly warmer on average. Relative humidity (RH2M) shows an inverse seasonal trend, with higher values in winter and lower in summer, reinforcing typical atmospheric moisture behavior.

Precipitation patterns (PRECTOTCORR) are highly skewed, with most days seeing little to no rainfall, punctuated by infrequent but significant events. Riga experiences slightly higher extremes compared to the others. Wind direction and speed data indicate that prevailing winds in all three cities predominantly originate from the southwest, with Riga experiencing the most intense and frequent winds—highlighting its suitability for wind energy deployment.

Solar radiation (ALLSKY_SFC_SW_DWN) displays the strongest seasonal variation. All three cities show solar peaks in June and July, with practically negligible radiation during winter months—confirming the need for seasonal energy planning strategies. Additionally, monthly solar distribution analysis affirms that summer months hold the most potential for solar infrastructure deployment, while winter months demand supplementary energy sources.

Surface pressure (PS) remains relatively stable, with Tallinn showing the widest pressure range and slightly higher medians than Vilnius. Although less directly tied to renewable energy, pressure dynamics contribute to understanding broader weather systems and planning considerations.

Overall, this analysis confirms that the Baltic region is characterized by strong seasonality in key weather variables. Summer months offer high solar potential, while winter seasons are better suited for wind exploitation. These insights establish a robust environmental foundation for regional energy planning and help contextualize the operational constraints of renewable systems in urban settings.

Responsible Data Guiding Principles

Our project investigates how local meteorological conditions influence renewable energy potential in Vilnius, Riga, and Tallinn. In conducting this analysis, we reflect on the six Responsible Data Guiding Principles to ensure that our data practices are inclusive, context-aware, and socially responsible.

1. Rethink Binaries

Our dataset comprises primarily numerical meteorological variables that may appear neutral at first glance. However, binary assumptions still emerge—such as sunny vs. Cloudy or summer vs. Winter, that oversimplify the atmospheric complexity and the spectrum of renewable energy performance. We challenge this binary framing by retaining "edge" meteorological cases, such as days with volatile solar radiation or mixed wind profiles, rather than removing them as outliers. This allows us to model a more nuanced view of environmental variability, acknowledging that energy systems operate under a continuum of weather conditions rather than simplified thresholds.

2. Embrace Pluralism

We recognize that data interpretation is not objective or universal, it is shaped by the perspectives of those who collect, analyze, and visualize it. Although our dataset is derived from NASA’s satellite data, which lacks socio-cultural context, we strive to incorporate plural viewpoints. For instance, while modeling solar radiation, we acknowledge that regional variations exist between Vilnius, Riga, and Tallinn. We also note the absence of urban socio-economic factors, and recommend future integration of stakeholder perspectives, such as from local energy cooperatives, urban planners, or residents affected by urban energy policy, into data selection and model interpretation.

3. Examine Power and Aspire to Empowerment

Meteorological data is often seen as neutral, but decisions about what data is collected (and what is left out) reflect underlying power structures. Our reliance on satellite data privileges remote sensing technologies over ground-based or community-collected data, which might better reflect local microclimates or neighborhood-specific concerns (e.g., urban heat islands). By explicitly acknowledging this imbalance, we aspire to create space for more participatory and community-embedded data practices in future iterations. Our predictive models are ultimately intended to support local governments and city planners in making informed, equitable, and sustainable decisions thus returning the value of the analysis to the communities it affects.

4. Consider Context

We actively consider the geographic and environmental context of the Baltic region. While our models are built on climate data over the time span of 5 years, we critically reflect on what is lost when ignoring urban infrastructure, energy access, or socio-economic disparities. For instance, a model might predict high solar potential, but without understanding building rooftop availability or energy poverty levels of the area, the insight is incomplete. We therefore stress that our models should be read within the meteorological scope they cover, not as comprehensive blueprints for urban policy, but as one layer in a broader interdisciplinary analysis.

5. Legitimize Embodiment and Affect

Traditional data analytics often prioritize numerical precision over human experience. Yet urban sustainability is not only a technical issue. It’s felt in heatwaves, cold spells, and infrastructure vulnerabilities. Though our models do not yet integrate embodied experiences (like heat stress or air quality discomfort), we recognize their importance. In the future, affective data, such as crowd-sourced perceptions of climate impacts or energy access disparities, could enrich our understanding and contribute to a more holistic vision of sustainability that centers lived experience.

6. Make Labour Visible

Finally, we highlight the often-invisible labor that underpins our work. From the engineers maintaining NASA’s POWER platform to the team preprocessing, cleaning, and modeling the data, many forms of effort support this analysis. Within our own team, we made roles and contributions explicit to ensure fair attribution. We also advocate for better documentation of data provenance, recognizing the human and institutional forces involved in shaping the datasets we rely upon.

3.Diagnostic Analytics:

How do solar, temperature, moisture, and wind variables correlate and interact?

Initially, we generated correlation heatmaps for each of the three cities individually. Subsequently, a correlation heatmap was produced for the merged dataset encompassing all three cities. Finally, we visualized the correlations for a subset of variables, selected based on our prior analysis.

The correlation matrix (see Fig. 9) provides valuable insight into the relationships among meteorological variables in our dataset. It shows strong positive correlations among solar radiation and temperature variables, while specific humidity also closely follows temperature patterns. Relative humidity is strongly negatively correlated with solar radiation and temperature. Wind speed metrics are highly interrelated, surface pressure shows weak correlations, and precipitation has only mild links to humidity and solar radiation.The thorough explanation of the variables can be found in the Jupyter Notebook.

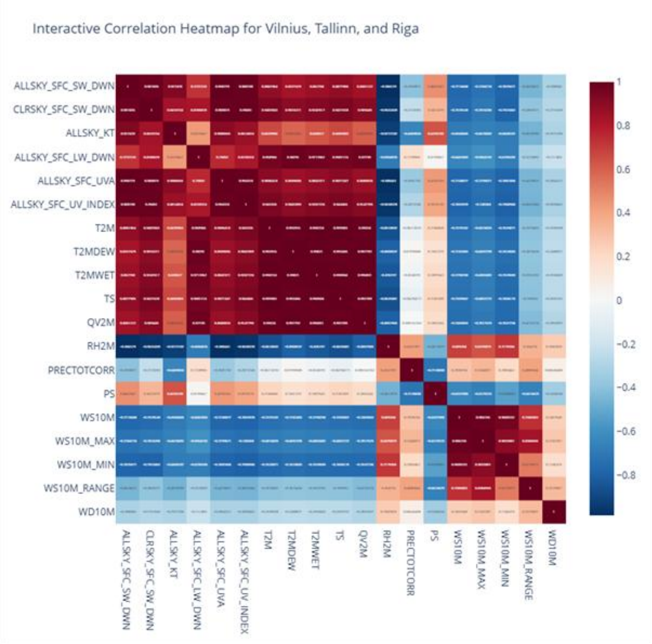

Fig. 9. Correlation heatmap of meteorological variables for Vilnius, Tallinn, and Riga.

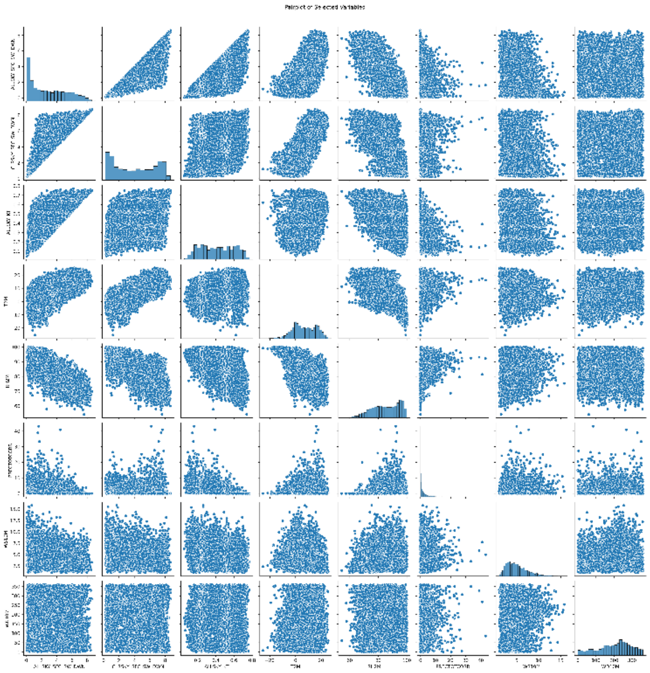

Fig. 10. Pairplot showing relationships among selected meteorological variables.

Fig. 10 visualizes the relationships among the selected meteorological variables. Each diagonal element displays the distribution (histogram) of a single variable, while each off-diagonal cell represents a scatter plot comparing two variables. Notably, strong linear patterns are visible between ALLSKY_SFC_SW_DWN (total shortwave radiation under all-sky conditions) and variables like ALLSKY_KT (clearness index) and CLRSKY_SFC_SW_DWN (clear-sky radiation), indicating high correlations. While these strong relationships may be useful for prediction, they also raise concerns about data leakage if any of these variables are derived from or directly influenced by the target. In contrast, variables such as WD10M (wind direction at 10m) and WS10M (wind speed at 10m) exhibit more scattered patterns, suggesting weaker or no clear correlation with the target.

Overall, this pairplot serves as a valuable tool for exploratory data analysis, helping to uncover feature redundancy, detect strong linear or nonlinear associations, and assess potential multicollinearity among predictors. It guides the feature selection process by highlighting which variables may offer unique information and which may overlap significantly with others. This understanding is essential for building robust and interpretable models, especially in time series or regression tasks where overfitting and multicollinearity can compromise model performance and generalizability.

Fig. 11 visualizes the relationship between solar radiation (ALLSKY_SFC_SW_DWN), air temperature at 2 meters (T2M), and wind speed at 10 meters (WS10M), with color representing relative humidity (RH2M). The plot reveals that higher solar radiation levels generally occur at higher temperatures and moderate wind speeds, while lower humidity levels are associated with increased solar irradiance. As humidity rises (indicated by the transition from blue to red in the color scale), solar radiation tends to decrease, suggesting an inverse relationship. The spatial clustering of data points illustrates how these meteorological variables interact, highlighting optimal conditions for solar energy potential.

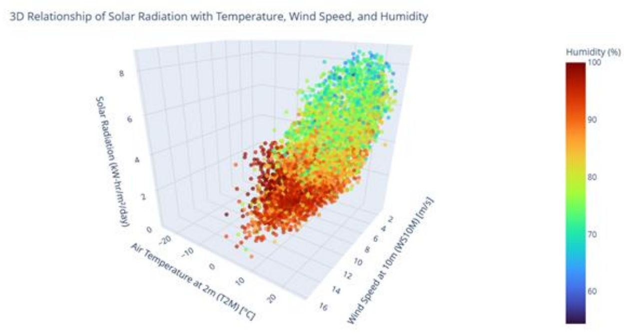

Fig. 11. 3D plot of solar radiation vs. temperature (T2M) and wind speed (WS10M), colored by humidity (RH2M).

Fig. 12 illustrates the daily trends of temperature (in red, °C) and humidity (in blue, %) from January 2018 to December 2022. Temperature exhibits a clear seasonal pattern, peaking in mid-year summers and dipping in winter, with extreme lows below -20°C observed particularly during the 2020–2021 winter. The average temperature line is shown as approximately 2.0°C. In contrast, humidity remains consistently high throughout the year, mostly ranging between

70% and 100%, with occasional dips likely corresponding to warmer periods. The dual y-axes effectively highlight the inverse relationship between temperature and humidity—higher humidity during cooler months and slightly reduced levels during warmer months—reflecting typical climate dynamics in temperate regions.

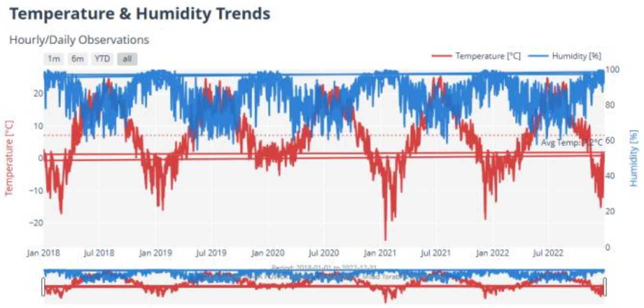

Fig. 12. Daily trends of temperature and humidity

Fig. 13 illustrates the seasonal relationship between air temperature (°C) and solar irradiance (kW-hr/m²/day) across Vilnius, Tallinn, and Riga. Each subplot corresponds to a city, and the data points are color-coded by season. Across all three cities, a clear seasonal pattern emerges:

- Summer (green) shows the highest solar irradiance, particularly at warmer temperatures (10°C to 25°C), reflecting longer, sunnier days.
- Spring (red) follows with moderately high radiation values at milder temperatures, indicating the seasonal transition.
- Autumn (purple) exhibits lower irradiance compared to spring despite similar temperature ranges, likely due to shorter daylight hours.
- Winter (blue) consistently displays both low temperatures and minimal solar radiation, as expected from the coldest, darkest months.

The overall distribution highlights how seasonality significantly influences the amount of solar energy received, even when temperatures overlap — with summer delivering the strongest irradiance at similar or even lower temperatures than spring or autumn. This plot is useful for identifying the best seasons for solar energy potential in each city.

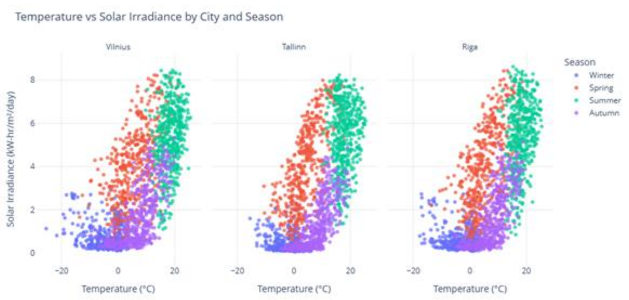

Fig. 13. the seasonal relationship between air temperature and solar irradiance.

Fig. 14 shows the seasonal trends of three solar-related variables (ALLSKY_SFC_SW_DWN, CLRSKY_SFC_SW_DWN, and ALLSKY_KT) across Vilnius, Tallinn, and Riga from 2018 to 2022. A dip around 2020–2021 is visible, followed by a rising trend, reflecting changes in atmospheric conditions and solar availability. The thorough explanation of the variables can be found in the Jupyter Notebook.

Fig. 14. The seasonal trends of three solar-related variables(ALLSKY_SFC_SW_DWN, CLRSKY_SFC_SW_DWN, and ALLSKY_KT)

Overall Insights:

Across all three Baltic capitals, the data reveals consistent seasonal and interannual variability in solar irradiance, with a clear increase during the summer months and near-zero values in winter. Notably, 2022 shows an upward trend across all variables, potentially indicating improved solar resource availability in the region. Among the three cities, Vilnius consistently receives slightly higher levels of solar irradiance, while Tallinn records the lowest, likely due to its more northerly latitude and prevailing weather conditions that reduce sunlight exposure.

To conclude, our diagnostic analysis reveals a strong, coherent structure among key meteorological variables that aligns well with established atmospheric science. Solar radiation is highly correlated with other solar-related metrics and shows a positive association with air temperature, particularly during summer months when irradiance is at its peak. Temperature and specific humidity (QV2M) are closely linked, reflecting the physical relationship that warmer air holds more moisture. In contrast, relative humidity (RH2M) exhibits a strong inverse correlation with both solar radiation and temperature, indicating that sunnier and warmer conditions typically correspond to lower relative humidity levels. Wind speed metrics are internally consistent but show weaker direct associations with solar radiation or temperature, though moderate wind speeds often accompany higher irradiance levels. These inter-variable dynamics underscore the need for integrated modeling approaches in urban energy planning, where solar, thermal, and moisture factors jointly influence the potential and reliability of renewable energy systems.

4.Predictive Analytics

Can we predict solar radiation based on other meteorological data that impact renewable energy generation and potential?

Now that we've explored trends and investigated variable interactions, we'll move into the prediction and forecasting phase. This step focuses on building models to estimate or forecast solar radiation, specifically the ALLSKY_SFC_SW_DWN variable—which represents the total shortwave (SW) radiation received at the surface under all-sky conditions (including clouds). To approach this, we can:

- 1.Use Generic Regression ("Prediction"): When we want a single-day estimate based on today's measurements (no explicit multi-step horizon).
- 2.Use Time-Series Forecasting ("Forecast"): When we explicitly model temporal dynamics to predict future values of ALLSKY_SFC_SW_DWN over a horizon (e.g., the next 7 or 30 days).

To evaluate whether solar radiation can be effectively predicted using meteorological data relevant to renewable energy systems, we designed a forecasting pipeline rooted in responsible and robust machine learning practices. The first step in this process involved selecting an appropriate forecasting horizon. While daily predictions are beneficial for short-term operational planning and are well-supported by the dataset's temporal resolution, they may not be optimal for long-term strategic forecasting, where weekly or monthly aggregation would be more effective. For the scope of this study, we focused on daily solar radiation prediction across a one-year horizon, using the full range of daily meteorological measurements as input.

Our next step involved selecting suitable regression models that could handle multivariate, nonlinear relationships inherent in environmental systems. We chose to implement a stacked ensemble approach that combines four diverse regression models: Random Forest Regressor, Gradient Boosting Regressor, Support Vector Regressor (SVR), and a Multi-Layer Perceptron (MLP). This ensemble strategy allows us to leverage the strengths of different algorithms, tree-based models for variance reduction, boosting for bias minimization, kernel methods for smooth function approximation, and neural networks for complex pattern recognition.

To ensure robustness and avoid overfitting, we employed time series–aware cross-validation(see Table. 2). The training and validation steps were performed using meteorological data from 2018 to 2021, while data from 2022 was reserved strictly for final testing. We trained three separate ensembles one for each city (Vilnius, Riga, and Tallinn) to ensure that models learned the specific climate dynamics of each urban environment without cross-pollinating information between them. Feature engineering was also applied to enhance temporal patterns and integrate time-series characteristics into the models.

Fold	Train start → end	Train size	Validation start → end	Val size
0	2018-01-01 → 2019-01-04	369 days	2019-01-05 → 2019-07-05	182 days
1	2018-01-01 → 2019-07-05	551 days	2019-07-06 → 2020-01-03	182 days
2	2018-01-01 → 2020-01-03	733 days	2020-01-04 → 2020-07-03	182 days
3	2018-01-01 → 2020-07-03	915 days	2020-07-04 → 2021-01-01	182 days
4	2018-01-01 → 2020-01-01	1097 days	2021-01-02 → 2021-07-02	182 days
5	2018-01-01 → 2021-07-02	1279 days	2021-07-03 → 2021-12-31	182 days

Table. 2. time series–aware cross-validation

A critical advantage of the stacking approach is the second-layer meta-model, which was implemented using RidgeCV regression. This meta-model learns to optimally combine predictions from the base models, down-weighting weaker or redundant contributions. For models requiring standardized input, such as SVR and MLP, data preprocessing was encapsulated within individual model pipelines using StandardScaler(), maintaining proper data flow hygiene throughout the ensemble architecture.

This stacking framework successfully blends the complementary strengths of diverse learners. Random Forest and Gradient Boosting handle nonlinearities and interactions among features, SVR adds robustness in smoother regimes, and MLP captures deep feature relationships. The ensemble's meta-learner, RidgeCV, ensures a regularized combination that reduces overall error while preserving interpretability.

Although this project did not extend to long-term or seasonal forecasting, we acknowledge the potential of more advanced techniques for future work. Classical time series models such as ARIMA or SARIMA could be applied for univariate prediction of solar radiation trends, while deep learning architectures like Long Short-Term Memory (LSTM) networks or Transformer models are well-suited for capturing long-term temporal dependencies in sequential data. Due to time constraints, these methods remain outside the current project's scope but represent promising avenues for future exploration.

Results explanations and Visualizations

Below is a consolidated summary table(Table. 3) of the models results, showing for each city the overall CV RMSE (mean ± std), and the final 2022 test metrics (RMSE, MAE, R^2):

Model Performance Summary

City	CV RMSE (mean ± std)	Test RMSE	Test MAE	Test R^2
Vilnius	0.870 ± 0.333	0.585	0.422	0.933
Tallinn	0.758 ± 0.174	0.658	0.440	0.925
Riga	0.837 ± 0.329	0.646	0.459	0.925

Table. 3. The summary table of the model's results

The fold-wise CV RMSEs and Test set results

As can be seen in the Table. 4, across all three cities, our stacking ensemble demonstrates strong and consistent performance both in cross-validation (2018–2021) and on the held-out 2022 test sets. During CV, the root-mean-squared error (RMSE) averaged 0.87 kW-hr/m²/day in Vilnius, 0.76 in Tallinn, and 0.84 in Riga—reflecting day-to-day variability but always within about 1 kW-hr/m²/day of the true value. Fold-by-fold R^2 scores climbed from roughly 0.60 in the smallest training windows up to over 0.93 once a full year of data was seen, with mean R^2 values above 0.87 in every city. On the 2022 test data, RMSE fell further (0.58 for Vilnius, 0.66 for Tallinn, 0.65 for Riga) and R^2 rose to ~0.93 across the board, indicating excellent generalization. Mean absolute percentage errors (MAPE) on the test sets settled between 22 % and 25 %, comparable to or better than typical solar-forecasting benchmarks, and explained variance exceeded 0.92 in every city. In short, our model reliably captures seasonal cycles and daily fluctuations of solar radiation, with marginal performance differences between cities.

	City	Fold	RMSE	MSE	MAE	MAPE	R2	Explained_Var
0	Vilnius	1	1.568270	2.459471	1.227999	37.619940	0.593938	0.817257
1	Vilnius	2	0.832714	0.693413	0.615242	45.745705	0.828760	0.854892
2	Vilnius	3	0.908162	0.824758	0.683202	30.385962	0.837694	0.847247
3	Vilnius	4	0.620138	0.384571	0.425060	39.163852	0.917356	0.922224
4	Vilnius	5	0.727364	0.529058	0.557460	25.251861	0.891526	0.898577
5	Vilnius	6	0.564912	0.319125	0.427998	41.108140	0.926540	0.932790
6	Tallinn	1	1.087107	1.181802	0.816912	45.109520	0.817476	0.847549
7	Tallinn	2	0.847382	0.718055	0.596652	116.492093	0.842723	0.864722
8	Tallinn	3	0.696206	0.484703	0.508973	40.606250	0.928260	0.928740
9	Tallinn	4	0.532057	0.283085	0.368267	34.644370	0.939871	0.939882
10	Tallinn	5	0.672289	0.451972	0.511349	37.207008	0.926902	0.937324
11	Tallinn	6	0.710412	0.504685	0.523133	60.665872	0.888187	0.929404
12	Riga	1	1.541131	2.375086	1.141712	39.285894	0.627587	0.721104
13	Riga	2	0.706875	0.499672	0.517846	56.733062	0.884581	0.887076
14	Riga	3	0.880653	0.775550	0.642800	28.985125	0.866948	0.879544
15	Riga	4	0.584517	0.341661	0.443130	62.380187	0.922818	0.934995
16	Riga	5	0.690024	0.476133	0.496183	23.799310	0.913913	0.914296
17	Riga	6	0.618037	0.381970	0.459477	49.780071	0.911400	0.938910
18	Vilnius	Test set	0.584721	0.341899	0.421547	25.582210	0.933146	0.933522
19	Tallinn	Test set	0.657837	0.432749	0.440240	22.111028	0.924941	0.926290
20	Riga	Test set	0.645535	0.416715	0.458544	24.303519	0.924753	0.924982

Table. 4 . The fold-wise CV RMSEs and Test set results

City-wise analysis:

Vilnius: In Vilnius(Fig. 15), we see a similarly tight fit: the stacked ensemble reproduces the characteristic summer plateau of high solar radiation and the jagged daily swings caused by passing weather fronts. While there are a handful of days (notably in late spring and early autumn) where the model slightly underestimates brief spikes, the predicted curve (orange dots) remains within about ±1 kW-hr/m²/day of the true values almost everywhere. This consistency confirms that seasonal features plus meteorological inputs give the model robust generalization for Vilnius's solar profile.

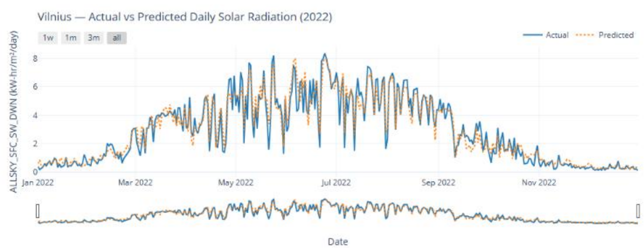

Fig. 15. The predicted value of solar radiation and the actual value

Tallinn: In Tallinn(Fig. 16), the model closely tracks the seasonal rise and fall of daily solar radiation, capturing the steep increase from near-zero winter values in January up to summer peaks around 8 kW-hr/m²/day in June and July, then gracefully following the autumn decline back toward winter lows. Small timing offsets appear on some cloudy days (e.g. under- or over-predictions by ~0.5–1 kW-hr/m²/day), but overall the dotted orange forecast line hugs the blue actual line throughout the year, demonstrating both strong seasonal learning and day-to-day variability in the model's predictions.

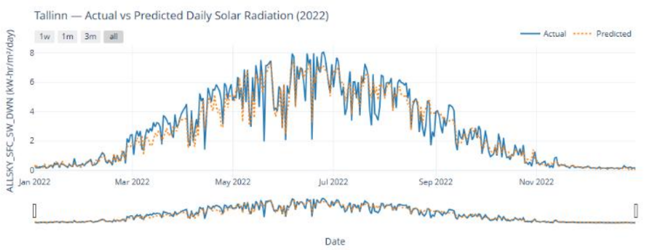

Fig. 16. The predicted value of solar radiation and the actual value

Riga: For Riga(Fig. 17), the forecast again aligns strongly with reality: winter values near zero, a clear ascent through spring, and a broad summer maximum around 7–8 kW-hr/m²/day. The model handles Riga's midseason variability well, though you can spot occasional underpredictions during sudden clear-sky days in May and slight overpredictions during intermittent cloud cover in late summer. Overall, the forecasted (dotted) and actual (solid) lines overlap so tightly that even the brief dips and peaks are faithfully mirrored, underscoring the model's ability to learn local seasonal and daily patterns.

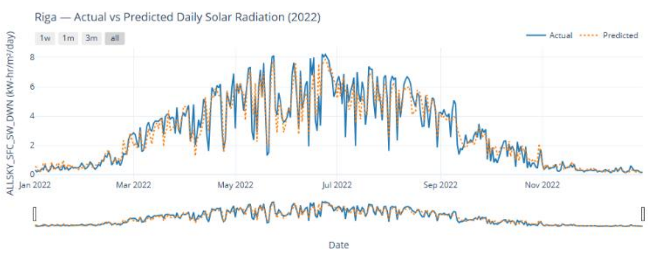

Fig. 17. The predicted value of solar radiation and the actual value

Fig. 18, six-panel "calendar" comparison shows how well our stacking model's day-ahead forecasts (bottom of each city block) track the true daily solar irradiance (top of each block) across Vilnius, Tallinn and Riga in 2022. In each city you can see the familiar summer peak (deep reds in June–July) and winter trough (pale yellows in December–January), and the model captures that seasonal rise and fall. However, the predicted heatmaps are generally a bit cooler and smoother—our forecasts tend to understate the very highest radiation days and slightly overstate the lowest ones—so the color gradients are more muted. Overall, though, the timing and relative patterns align closely, demonstrating that the model reliably learns the shape of the solar cycle in all three locations.

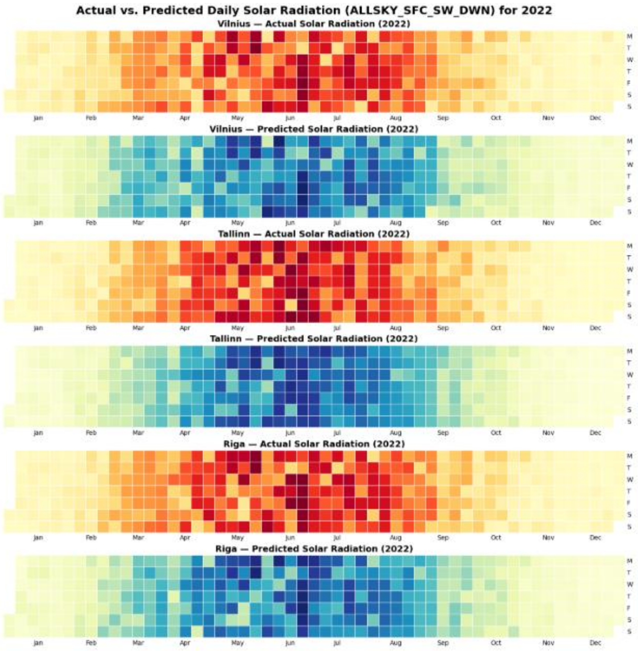

Fig. 18. Six-panel "calendar" comparison shows the stacking model's day-ahead forecasts

SHAP: Post-Hoc Explainability

To explain the output of our predictive model, we use SHAP (SHapley Additive exPlanations) values. SHAP allows us to quantify the contribution of each feature to an individual prediction, making the model's decisions more transparent. This approach helps both technical users and non-experts understand why the model made a specific prediction by revealing the underlying reasoning in a human-interpretable way.

Below, we implement the SHAP analysis based on the following steps:

- Step 1: load the dataset and Train the model (to fit SHAP input format)(we already have done this step)
- Step 2: Create and Instentate a SHAP explainer (KernelExplainer)
- Step 3: Bar chart of mean feature importance (Global Explanations of feature importance)
- Step 4: Visualize a single prediction(s) (Local Explanation)
- Step 5: Visualize many predictions (Local Overview) or Filtered predictions (Cohort)
- Step 6: Summary Plot

Important Note: In the scope of this project, SHAP analysis was performed only for the Vilnius dataset. Ideally, the same interpretability analysis should be applied to the other two cities (Tallinn and Riga) to ensure consistency and fairness in model evaluation. However, due to limited time and computational resources, we have skipped those for now.

Bar chart of mean feature importance (Global Explanation): Fig. 19 shows that **Relative Humidity (RH2M)** is by far the strongest predictor—on average, a one-unit change in RH2M shifts the solar irradiance estimate by about 0.76 kW·hr/m²/day—followed by **DayOfYear** (seasonality), **Surface Temperature (TS)**, and **Precipitation**. Air temperature at 2 m, week-of-year, and a small year-on-year trend also matter, while wind-related features and pressure have almost no impact on average. In Vilnius, moisture and seasonal timing dominate the model's decisions, reflecting how cloud cover and the sun's annual cycle drive daily solar radiation.

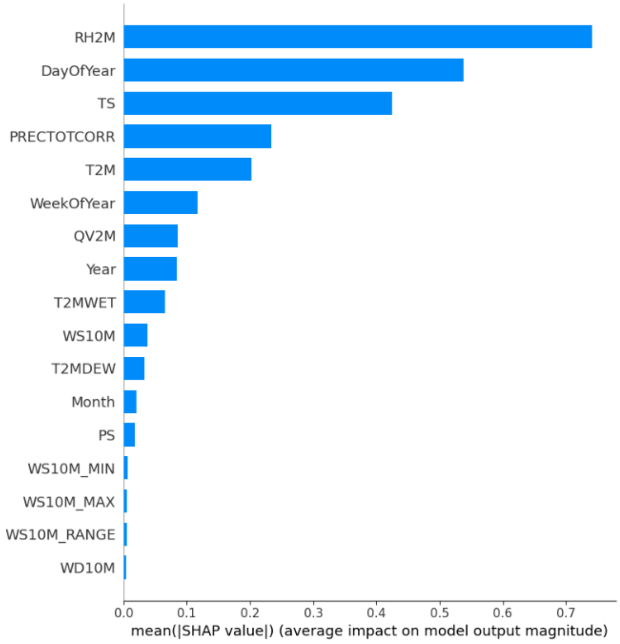

Fig. 19. The bar chart of mean feature importance

Visualize a single prediction (Local Explanation): The **waterfall/force plot**(Fig. 20) zooms in on one particular April day: it starts at the model's baseline prediction (≈2.85 kW·hr/m²), and shows in red how RH2M, DayOfYear, TS, etc., each "push" the estimate up or down (blue) to the final output (≈5.32 kW·hr/m²). For that single case, high humidity and seasonal timing add the largest positive contributions, skin temperature subtracts a bit, and the remaining features nudge the forecast very slightly. Together, these charts give both a bird's-eye view of feature importance and a transparent, step-by-step breakdown of individual predictions.

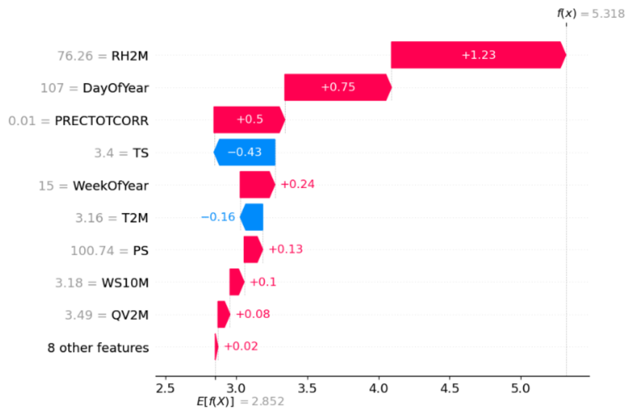

Fig. 20. The Waterfall plot indicating Single prediction (Local Explanation)

In Fig. 21, we see 100 individual force-plot "lines," each one showing how the same ordered features cumulatively move the baseline output (≈2.85) up or down to each day's prediction. Stacked red bands indicate features boosting the forecast, blue bands show dampening effects, and the vertical axis orders days by similarity. we can spot clusters—e.g. many lines dip in the middle when humidity is low (blue), then rise again when seasonal and humidity effects combine—revealing consistent behavior across the time series and highlighting outlier days whose feature patterns differ markedly.

Fig. 21 . The representation of 100 individual force-plot

SHAP Summary Plot: As depicted in Fig. 22, each dot in this plot represents one of our 100 test-day explanations for a given feature. The horizontal position shows whether that feature pushed the model prediction higher (positive SHAP) or lower (negative SHAP), and the color encodes the actual feature value (red = high, blue = low). For example, for RH2M, high-humidity days (red) almost always drive large positive contributions (dots far right), whereas low-humidity days (blue) tend to push the prediction down. The wide horizontal spread for RH2M and DayOfYear confirms they're the most influential inputs, while features near the bottom (e.g. WS10M_MAX, WD10M) cluster tightly around zero, indicating negligible impact across all samples.

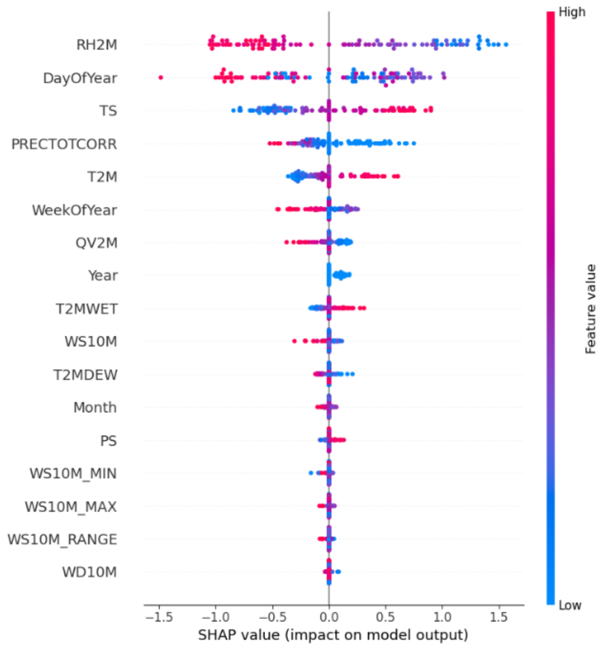

Fig. 22 . The SHAP Summary Plot

In conclusion, our analysis confirms that solar radiation can indeed be effectively predicted using other meteorological variables that influence renewable energy generation and potential. By integrating temperature, humidity, wind speed, and pressure data into a robust stacked ensemble modeling framework, we were able to generate accurate and generalizable daily predictions of solar radiation across three distinct urban climates. The strong correlations observed among key atmospheric variables, combined with the diversity and complementarity of our modeling techniques, underscore the viability of data-driven approaches in forecasting solar potential. These predictive capabilities are not only technically sound but also highly relevant for informing short-term energy planning, optimizing renewable resource allocation, and supporting data-informed urban sustainability strategies in the Baltic region and beyond.

5.Prescriptive Analytics

Turning Solar Forecasts into Actionable Decisions

With accurate day-ahead and intra-day solar irradiance predictions in hand, decision-makers can move from "what will happen" to "what should we do." Based on the model's predictions of solar radiation, we can take action to improve energy efficiency and better prepare for future conditions. The model tells us how much solar energy is likely to be available on any given day, allowing energy planners to make smarter decisions. These include deciding when to charge batteries, when to use stored power, and when to reduce energy consumption during lower sunlight hours. By understanding what factors impact solar power—like humidity, temperature, or cloudiness—we can make better plans for clean energy use and reduce energy waste. Below are some prescriptions informed by our stacking model:

Maintenance & Cleaning Operations

- Panel Washing: Schedule panel cleaning on days with poor forecast (e.g. heavy cloudiness or precipitation) to minimize opportunity-cost of downtime. For instance, target days where predicted irradiance is below the 25th percentile.
- Inspection Windows: Allocate maintenance crews on low-yield days (when SHAP shows humidity and precipitation strongly depress output) to maximize solar plant availability during peak-yield periods.

More technical actions to take:

Optimal PV Generation Scheduling

- Dispatch Battery Storage: When tomorrow's forecasted irradiance exceeds a high threshold (e.g. > 5 kW·hr/m²), postpone discharging stored energy until late afternoon—maximizing self-consumption of abundant solar and avoiding grid peak charges. Conversely, on low-irradiance days (< 2 kW·hr/m²), pre-charge batteries during overnight off-peak hours to ensure reliable supply.
- Curtailment & Grid Export: If local generation is forecast to exceed on-site load by > 20%, schedule controlled curtailment or negotiate feed-in tariffs in advance to avoid inverter clipping and lost revenue.

Demand-Side Management & Load Shifting

- Industrial Processes: For operations that can be time-shifted (e.g. water pumping, electrolysis), run energy-intensive tasks during high-irradiance windows identified by the model (e.g. DayOfYear 120–240), reducing reliance on grid power and cutting costs.
- Residential Smart Charging: Integrate predictions into EV-charging schedules: ramp up home chargers or vehicle-to-grid discharge when forecast predicts surplus solar, and throttle back during low-solar hours.

Financial Hedging & Market Bidding

- Day-Ahead Market Offers: Use irradiance forecasts to fine-tune day-ahead generation bids, reducing imbalance penalties. For example, bid conservatively on days where high relative humidity signals cloudy conditions.
- Option Strategies: Hedge against low-solar risk by purchasing call options on peak-demand hours when forecasts indicate under-production relative to baseline.

What-If Analysis chart

What If?	Consequences	Likelihood	Severity	Recommendations
Humidity spikes for a full week	Less solar energy is produced. Daily output drops.	Medium	Moderate	Use stored power wisely. Schedule high-energy tasks earlier in the day.
Several clear sunny days in a row	High solar energy generation, more than average.	High	Low	Charge storage systems fully. Run energy-heavy tasks during peak sun hours.
Unusual afternoon cloud coverage	Sudden drop in energy during the afternoon peak.	High	High	Shift key activities to mornings. Use backup power sources to meet energy needs.

How the model will be used in real life

In a real-world setting, this stacking-ensemble solar-radiation model would sit at the heart of an automated daily forecasting pipeline that drives decision-making across generation, operations and planning:

- **Data Ingestion & Preprocessing**
  • Every morning (or more frequently), the meteorological data feed (the same 19 GHI/temperature/humidity/wind parameters) is automatically pulled and time-indexed.
  • A small ETL job applies exactly the same feature-engineering steps we used in development (dropping leakage features, computing Year/DayOfYear/WeekOfYear, scaling for SVR/MLP pipelines, etc.).
- **Prediction Service**
  • A lightweight API (e.g. Flask/FastAPI or an enterprise scheduler) loads the latest cleaned features and calls our pre-trained stacking model for Vilnius, Tallinn and Riga.
  • It writes out a day-ahead forecast of ALLSKY_SFC_SW_DWN for each location—for example, as a simple CSV, database table, or JSON payload.
- **Integration with Operations**
  • PV Plant Scheduling: Operators use the forecast to size inverter output, schedule panel cleaning/maintenance on low-irradiance days, and optimize battery charge/discharge cycles.
  • Grid Balancing & Market Bids: Utilities and traders feed the day-ahead solar forecasts into their unit-commitment models to decide how much conventional generation to keep spinning and what appetite to bid into spot markets.
  • Demand Response & Storage Management: Energy managers adjust flexible loads (e.g. EV charging, HVAC pre-cooling) when high-radiation days are expected, maximizing consumption of clean solar.
- **Visualization & Alerts** • A daily dashboard shows side-by-side heat-maps (actual vs. predicted) and key error metrics.
  • Automated alerts (e.g. via Slack or SMS) flag any unusually low or high forecasts—so teams can investigate sensor outages or extreme weather risks.
- **Feedback & Retraining**
  • At the end of each month/quarter, the actual 2022 (and beyond) observations are compared against our forecasts to track model drift.
  • we schedule a quarterly retraining job that incorporates the newest data into our stack, re-optimizes hyperparameters (if needed), and refreshes the production model.
- **Governance & Continuous Improvement**
  • we log predictions, actuals and feature distributions to monitor data quality, model stability and bias (e.g. does it under-predict in winter?).
  • If performance dips below a threshold (say RMSE > 1 kW-hr/m²), a notification triggers a deeper error-analysis process or even an automated rollback to the last known-good model.

By embedding this model into a repeatable, monitored forecasting pipeline—complete with automated data ingestion, model serving, dashboarding and retraining— we turn our proof-of-concept into a robust, production-grade decision-support system that helps PV operators, grid planners and energy traders make cleaner, more efficient, data-driven choices every single day.

In this chapter, we have seen that accurate solar radiation forecasting enables a shift from reactive to proactive energy management, allowing for both strategic and operational optimization of renewable energy deployment. At a general level, strategies include aligning energy consumption with periods of high solar availability, integrating forecasts into battery storage and dispatch systems, and adjusting infrastructure maintenance to minimize production losses. Predictive insights also support real-time load balancing, smart charging of electric vehicles, and the strategic use of backup power during low-output periods. These practices not only increase the efficiency of solar systems but also contribute to broader climate and sustainability goals turning predictive analytics into a cornerstone of smarter, cleaner energy systems.

Conclusion

This project systematically explored how variations in meteorological conditions impact renewable energy potential and urban sustainability in the Baltic capitals of Vilnius, Riga, and Tallinn. Leveraging five years of detailed meteorological data from NASA's POWER platform and employing responsible data analytics, we addressed the research question: How do variations in meteorological parameters influence renewable energy potential and urban sustainability in Vilnius, Tallinn, and Riga?

The analysis revealed that meteorological parameters strongly influence both the feasibility of renewable energy systems and the broader sustainability of urban environments.

Solar Radiation showed distinct seasonal peaks during the summer months, indicating high potential for solar energy generation from May to August. This pattern supports the strategic deployment and scaling of photovoltaic infrastructure in these months.

Temperature plays a key role in shaping urban energy demands, especially for heating and cooling systems. Warmer periods correspond with increased solar potential but also influence consumption patterns that need to be managed for optimal energy efficiency.

Wind Conditions, particularly the consistent winds from southern and western directions, offer strong opportunities for wind energy, especially in Riga, where wind intensity and frequency are highest among the three cities.

These findings highlight the importance of tailored, city-specific renewable energy strategies that are responsive to local climatic patterns and inter-variable relationships. The use of descriptive, diagnostic, predictive and prescriptive analytics enabled a better view of these dynamics. Descriptive analytics revealed consistent seasonal trends, notably that solar radiation peaked during summer, temperature variations significantly influenced energy consumption patterns, and prevailing winds primarily originated from southern and western directions. Diagnostic analytics highlighted robust correlations, particularly between temperature and humidity, and solar irradiance and temperature, underscoring the climatic factors crucial for renewable energy strategies.

The predictive analytics phase developed a robust stacked ensemble model leveraging RandomForest, GradientBoosting, SVR, and MLP, achieving precise predictions for solar radiation based on meteorological inputs. The model performance confirmed the suitability of machine learning approaches for forecasting renewable energy potential under varying weather conditions.

Our predictive analytics approach, using a stacked ensemble of RandomForest, GradientBoosting, SVR, and MLP, delivered robust and accurate predictions of solar radiation. The performance of this model validates the effectiveness of machine learning techniques for informing renewable energy planning, especially when guided by responsible and context-aware data practices.

Recommendations

To capitalize on our findings and advance sustainable urban development aligned with EU energy objectives, we offer a series of strategic recommendations grounded in meteorological insights and data-driven modeling.

Firstly, cities should implement seasonal energy strategies that align renewable energy deployment with environmental conditions. During the high-yield months of May through August, when solar irradiance reaches its peak, solar infrastructure should be prioritized to maximize energy generation efficiency. Conversely, winter months—characterized by diminished solar availability—present an opportunity to shift focus toward wind energy systems. This is particularly relevant for cities like Riga, which experience favorable wind patterns and could offset seasonal solar deficits through intensified wind energy utilization.

At the regional level, a differentiated approach to renewable energy optimization is essential. Vilnius, which consistently demonstrates slightly higher solar irradiance compared to its neighbors, is particularly well-suited for large-scale solar energy expansion. Riga, on the other hand, benefits from robust wind conditions and should emphasize the development of urban wind energy infrastructure, potentially serving as a regional model for integrated wind solutions. Tallinn's more balanced solar and wind profile suggests it would benefit most from a hybrid renewable system that combines both sources to provide energy resilience and operational flexibility.

From an urban planning and infrastructure perspective, integrating predictive meteorological modeling into city development frameworks can significantly enhance renewable energy siting decisions and guide long-term infrastructure investments. Local governments should also consider incorporating weather-based insights into the revision of building codes and urban design standards, thereby promoting neighborhood-level climate resilience and energy efficiency.

On the technical front, future phases of this work should expand the dataset to include socio-economic indicators, infrastructure data, and policy variables. Such an enriched dataset would enable a more holistic understanding of the drivers of urban sustainability. Additionally, the use of advanced machine learning models, including long short-term memory (LSTM) networks and transformer-based architectures, should be explored to support more robust long-term forecasting and strategic scenario planning.

Finally, we underscore the importance of continuous monitoring and adaptive feedback loops. Establishing dynamic data collection pipelines and retraining predictive models on an ongoing basis will ensure that planning tools remain relevant amid changing climate conditions and urban growth. Moreover, real-time monitoring of installed renewable systems should inform policy updates and infrastructure adjustments, allowing cities to adapt their energy strategies proactively.

These recommendations, rooted in the specific meteorological profiles of Vilnius, Riga, and Tallinn, and supported by predictive analytics, offer a roadmap for Baltic capitals to not only meet EU climate targets but also enhance their energy autonomy and urban resilience.

Reflection

Responsible Data, Fairness, Transparency, and Explainability

This project explored the renewable energy potential of the Baltic States through meteorological modeling. And has been shaped by a commitment to responsible data science. Our reflection weaves together essential considerations on privacy, positionality, fairness, transparency, and explainability, ensuring that the insights produced by our models are ethically grounded, contextually aware, and critically assessed.

Privacy, Ethics, and Future Risks

While our dataset does not include personally identifiable information, comprising satellite-derived variables like temperature, humidity, and solar radiation. Privacy remains a pertinent issue when envisioning future applications. Predictive models built from environmental data could eventually intersect with smart grid systems or real-time infrastructure monitoring. In such scenarios, even anonymized data, when paired with spatial or demographic overlays, may risk indirect privacy breaches or community-level surveillance.

As cities like Vilnius, Riga, and Tallinn transition to smarter infrastructures, data collection could become more granular and potentially invasive. Ethical data use, therefore, requires proactive design: clear communication of data provenance, rigorous anonymization practices, and caution against presenting environmental data as inherently neutral.

Positionality and Reflexivity

As a group of international students from diverse technical backgrounds, our worldview is informed by academic environments in the Global North. We are trained to value modeling and prediction, and this often shapes our framing of sustainability challenges in technocratic terms. However, this lens can obscure the lived experiences of communities that data alone cannot represent.

We acknowledge the epistemological limitations of our position and seek to mitigate this by naming our assumptions and the dataset's blind spots. Particularly the absence of socio-economic, cultural, or participatory inputs. Our goal is to evolve our methods toward a more inclusive, co-designed model of analytics that centers marginalized voices and local knowledge.

Fairness in Modeling and Prediction

Our approach to fairness draws from the concepts of both individual and group fairness, emphasizing not just equal treatment but equitable outcomes. In practice:

Representation bias arises from our exclusive focus on three urban centers, leaving rural and peri-urban populations out of scope.

Aggregation bias is partially addressed by training city-specific models, but without demographic or infrastructural diversity, sub-group fairness cannot be validated.

Deployment bias becomes critical if predictive insights are translated directly into infrastructure investments, potentially disadvantaging areas not captured by our model.

To counter this, we carried out a "what-if" analysis exploring the consequences of solar deployment based solely on high-prediction zones. We concluded that future implementation should incorporate socio-demographic overlays and equity-weighted policy criteria to prevent unjust outcomes.

Bias Analysis and Reflection

Historical bias refers to patterns in data that reflect and perpetuate historical inequalities, institutional decisions, or technological constraints. In our project, although we work with meteorological data that appears neutral on the surface, historical bias can still be subtly embedded within the dataset and its interpretation. The choice of data source: NASA's POWER dataset, reflects a reliance on global satellite infrastructure, which, while comprehensive, has its limitations. Historically, such datasets have been optimized for global consistency rather than localized accuracy, potentially underrepresenting microclimatic variations that are relevant for city-scale planning.

Furthermore, our exclusive focus on capital cities; Vilnius, Riga, and Tallinn reflects a broader historical pattern in which policy attention and infrastructural development are concentrated in political and economic hubs. This focus inadvertently reproduces a dynamic where peripheral and rural regions remain analytically invisible, potentially leading to skewed investment strategies and reinforcing long-standing disparities in resource distribution. Similarly, the five-year time window of the dataset (2018–2022) excludes longer-term climate trends and historic weather anomalies, limiting our understanding of variability and the effects of climate change over time.

Recognizing these historical patterns, our work takes care not to present its outputs as universally representative. We advocate for complementing satellite-derived datasets with ground-level observations and community-based knowledge, particularly in future iterations where implementation decisions may depend on highly localized insights. Addressing historical bias requires not only technical adjustments but also a broader shift in how we define data relevance and whose realities we choose to represent in predictive systems.

Representation bias arises when the data used in a project does not fully reflect the diversity of the subject population or geographic scope. In our analysis, this bias becomes evident through the narrow spatial focus of the dataset, which includes only the capital cities of Lithuania, Latvia, and Estonia. While these urban areas offer valuable insights due to their data availability and strategic importance, they do not capture the full meteorological or infrastructural landscape of their respective countries. As a result, the generalizations drawn from the models may not be transferable to suburban, rural, or coastal areas, where renewable energy potential and urban form may vary significantly.

Additionally, the dataset contains only meteorological variables, omitting crucial dimensions such as socio-economic status, energy access disparities, or local energy infrastructure. This restricts the model's representational capacity to environmental phenomena alone and neglects the social and economic contexts in which renewable energy policies are enacted. In particular, the absence of data on marginalized communities limits the ability to evaluate how renewable energy strategies might impact vulnerable populations differently.

To mitigate representation bias, we explicitly contextualize our predictions as geographically and thematically bounded. Future iterations of this project should prioritize expanding data inputs to include demographic and infrastructural data layers. Moreover, participatory approaches that engage local stakeholders in defining project goals and selecting relevant data can help ensure that the analytics serve diverse communities, not just the most visible or data-rich.

Measurement bias occurs when the variables selected to approximate a concept are inadequate, leading to distorted insights or misaligned interpretations. In our project, measurement bias stems primarily from the reliance on satellite-derived estimates of atmospheric conditions. Although NASA's POWER dataset is robust and well-regarded, the environmental variables are modeled outputs rather than direct measurements. Derived variables such as the clearness index or relative humidity may reflect average atmospheric conditions, but they may not accurately capture hyper-local conditions in dense urban environments, such as the urban heat island effect or wind behavior around tall buildings.

Moreover, our use of daily averages for features like temperature and solar irradiance introduces aggregation bias, which can obscure important intra-day variations that are critical for energy production modeling. For example, rooftop solar panel efficiency can vary greatly within a single day depending on cloud cover patterns and peak sunlight hours, details that daily averages cannot fully capture. Additionally, the lack of calibration with ground-truth meteorological stations means there is limited validation of the accuracy of the satellite-derived metrics for each city.

Another important dimension of measurement bias involves the absence of certain metrics that could enhance the model's realism. For example, while we predict solar radiation potential, we do not account for built environment features like building orientation, roof inclination, or shading all of which significantly affect actual energy output.

To reduce measurement bias in future work, we recommend integrating ground-sensor data where available, using higher-resolution temporal data (e.g., hourly rather than daily), and expanding the feature set to include physical infrastructure and land use patterns. These steps would enhance the model's ability to reflect the real-world complexity of renewable energy generation and deployment.

Transparency and Explainability

Transparency means revealing not only how the model predicts but also why it was built, how it processes data, and what risks it poses. Explainability, particularly through SHAP analysis, enabled us to unpack the influence of features like clearness index (ALLSKY_KT), temperature (T2M), and humidity (RH2M), both globally and locally.

These explanations helped stakeholders visualize how different weather patterns shape predictions. They also enhanced our team's ability to debug, validate, and communicate results. Yet we remain cautious: SHAP values show correlation, not causation, and can mislead when oversimplified. To ensure responsible interpretation, we contextualized SHAP outputs with traditional metrics and domain insights, emphasizing their limits and potential for misuse.

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